# ICT2122 · Object Oriented Programming

## Core Concepts: Inheritance and Polymorphism

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

# Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

# 1. Main idea

Inheritance expresses a subtype relationship only when the subtype can honor the parent contract. Polymorphism allows code to call an interface operation while the runtime object supplies the implementation. Composition instead assembles behavior from collaborating objects.

# 2. Example and reasoning

A payment workflow can depend on a PaymentMethod interface, with separate implementations for a demonstration card and bank transfer. The workflow remains independent of each method’s internal details.

# 3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

# 4. Questions for study

- Explain “core concepts: inheritance and polymorphism” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

## Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.